

Experiment No. 1

Study of DC Characteristics of an N-MOS Transistor

1. Aim

To design a test bench for an N-MOS transistor in Cadence Virtuoso using the GPD090 technology library, and to obtain and study its DC output characteristics (Drain Current I_D versus Drain-to-Source Voltage V_{DS}) for different values of Gate-to-Source Voltage (V_{GS}) using DC and Parametric analysis in ADE-L.

2. Theory

An N-channel MOSFET (NMOS) is a four-terminal semiconductor device Gate (G), Drain (D), Source (S) and Body/Substrate (B) fabricated on a p-type substrate with two heavily doped n+ regions that form the source and drain. A thin gate oxide separates the polysilicon/metal gate from the substrate, and the channel region lies between the two n+ regions, directly beneath the gate.

When a sufficiently positive voltage is applied at the gate with respect to the source ($V_{GS} > V_{TH}$, the threshold voltage), an inversion layer (n-type channel) is formed between the source and drain, allowing current to flow when a drain-to-source voltage (V_{DS}) is applied. Depending on the relative magnitudes of V_{GS} , V_{DS} and V_{TH} , the NMOS operates in one of three regions:

- Cut-off region: $V_{GS} < V_{TH}$ no channel is formed and the drain current $I_D \approx 0$.
- Triode / Linear region: $V_{GS} > V_{TH}$ and $V_{DS} < (V_{GS} - V_{TH})$ the device behaves like a voltage-controlled resistor and I_D increases roughly linearly with V_{DS} .
- Saturation region: $V_{GS} > V_{TH}$ and $V_{DS} \geq (V_{GS} - V_{TH})$ the channel is pinched off near the drain and I_D becomes almost independent of V_{DS} , depending mainly on V_{GS} .

The output (drain) characteristics of an NMOS are obtained by plotting I_D versus V_{DS} for several fixed values of V_{GS} . Each curve first rises through the triode region and then flattens out in saturation; curves for higher V_{GS} produce higher saturation currents, since the effective overdrive voltage ($V_{GS} - V_{TH}$) is larger.

3. Software / Tools Required

- Cadence Virtuoso Schematic Editor
- GPDk090 Technology / Analog Design Kit
- Analog Design Environment (ADE-L) for DC and Parametric Analysis

5. Circuit Diagram

The test bench consists of a single NMOS transistor (gpdk090_nmos1v, $W = 120\text{ nm}$, $L = 100\text{ nm}$) whose Gate is biased through a DC voltage source VGS, whose Drain is biased through a second DC voltage source VDS (swept from 0 V to 1.8 V), and whose Source and Body are tied to ground (gnd).

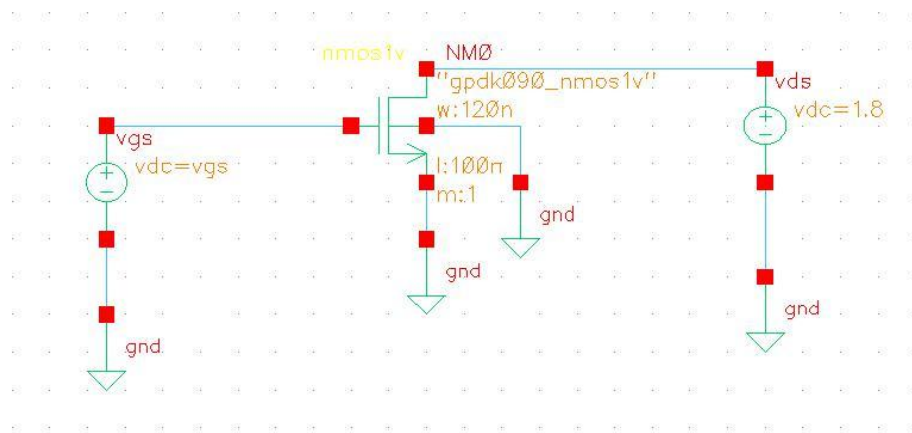


Fig. 2: Schematic of the NMOS DC characterization test bench built in Cadence Virtuoso

6. Procedure (Steps to Perform the Experiment)

6.1 Creating the Library and Schematic

1. Open Cadence Virtuoso. Go to File → New → Library, attach it to the GPDk090 technology library, and click OK.
2. Go to File → New → Cellview. Enter a suitable cell name (e.g., "Lab1_NMOS"), keep the tool as Schematic (Composer-Schematic), and click OK.
3. In the schematic window, go to Add → Instance (or the Add Instance icon). Click Browse and select the library "GPDk090".
4. From the cell list, select the cell "nmos1v" with View: symbol, and place it in the schematic window.
5. Add the Ground/Power symbols from the analogLib library place a "gnd" symbol and "vdc" (DC voltage source) symbols as required for VGS and VDS.
6. Wire the connections as shown in the circuit diagram (Fig. 2): Gate → VGS source, Drain → VDS source, Source and Body → gnd.
7. Check the schematic for errors (Check and Save) and save the design.

6.2 Setting up the DC Analysis (ADE-L)

8. Launch the Analog Design Environment: Launch → ADE-L.
9. Go to Analyses → Choose, and select the "dc" analysis type from the Analyses (AC / DC / Trans) options.
10. In the DC Analysis form, set Component Parameters: pick the Drain-side voltage source (VDS) as the component to sweep.
11. Set the sweep range: Start = 0 V, Stop = 1.8 V, with Sweep Type = Linear and Step Size = 0.2 V.
12. Under Outputs, select "To be plotted" and click on the desired node/terminal on the schematic (drain current through the transistor) to add it as an output — this generates the I_D vs V_{DS} plot (Graph 1).
13. Click on the DC Voltage field of the VGS source in the schematic to select it for the parametric sweep, and run the simulation to view the first output characteristic curve.

6.3 Parametric Analysis (Family of Curves for Different VGS)

14. Go to Tools → Parametric Analysis (Add Variables) in the ADE-L window.
15. Select the variable VGS, and set its Range = 0 V to 1.8 V, with Step Mode = Linear Steps and Step Size = 0.2 V.
16. Run the parametric sweep. This re-runs the DC analysis for each stepped value of VGS, producing a family of I_D – V_{DS} curves — one curve per VGS value — on the same graph (Graph 1).
17. Individual curves can also be isolated/viewed one at a time by toggling visibility in the plot legend (Graph 2).

7. Result / Observations

The DC output characteristics of the NMOS transistor were plotted with the drain current I_D (in μA) on the Y-axis and the drain-to-source voltage V_{DS} (in V) on the X-axis, for V_{GS} stepped from 0 V to 1.8 V in steps of 0.2 V.

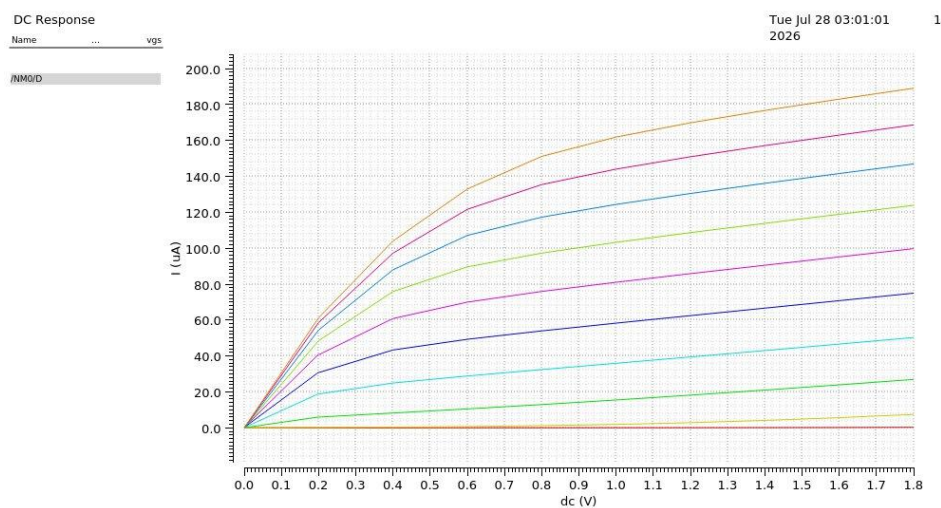


Fig. 3: Output characteristics (I_D vs V_{DS}) of the NMOS: family of curves for V_{GS} stepped from 0 V to 1.8 V (parametric analysis)

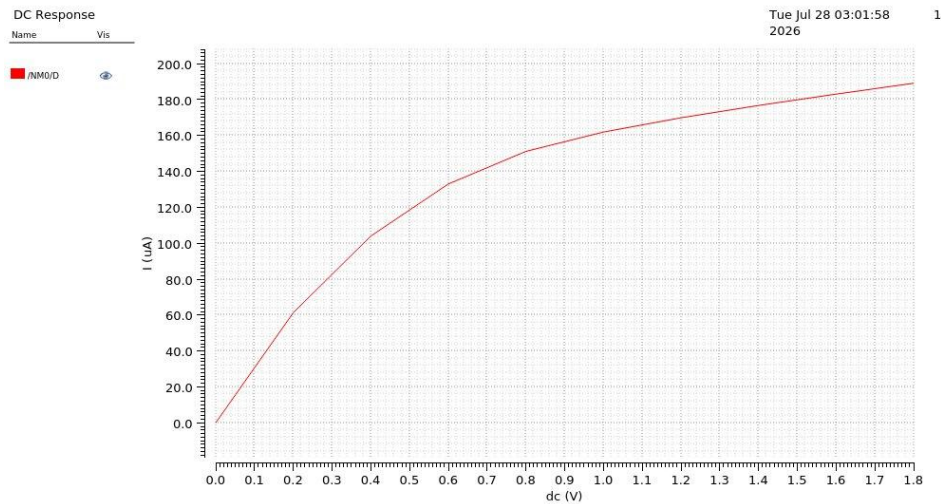


Fig. 4: Output characteristic (I_D vs V_{DS}) of the NMOS for a single fixed value of V_{GS} , isolated from the family of curves in Fig. 3

As seen in Fig. 3, each curve rises steeply for small V_{DS} (triode/linear region), then bends and flattens as V_{DS} increases beyond $(V_{GS} - V_{TH})$, entering the saturation region where I_D becomes nearly constant. Curves corresponding to higher V_{GS} values saturate at higher drain current levels, confirming the expected quadratic-like dependence of saturation current on the gate overdrive voltage.

8. Conclusion

The DC (output) characteristics of an N-MOS transistor were successfully obtained in Cadence Virtuoso using the GPDK090 technology by sweeping V_{DS} from 0 V to 1.8 V through DC analysis and stepping V_{GS} from 0 V to 1.8 V through parametric analysis. The resulting family of I_D - V_{DS} curves clearly shows the triode and saturation regions of operation and demonstrates that the drain current increases with increasing gate-to-source voltage, consistent with standard MOSFET theory. This experiment provided hands-on familiarity with schematic capture, DC analysis setup, and parametric sweeps in the Cadence ADE-L environment.

Experiment No. 2

Study of DC Characteristics of a P-MOS Transistor

1. Aim

To design a test bench for a P-MOS transistor in Cadence Virtuoso using the GPD090 technology library, and to obtain and study its DC output characteristics (Drain Current I_D versus Drain-to-Source Voltage V_{DS}) for different values of Gate-to-Source Voltage (V_{GS}) using DC and Parametric analysis in ADE-L.

2. Theory

A P-channel MOSFET (PMOS) is a four-terminal semiconductor device Gate (G), Drain (D), Source (S) and Body/Substrate (B) fabricated with two heavily doped p+ regions forming the source and drain inside an n-type substrate (or an n-well, in a standard CMOS process). A thin gate oxide separates the gate from the substrate, and the channel region lies between the two p+ regions, directly beneath the gate.

The PMOS conducts in the opposite manner to an NMOS: a sufficiently negative gate-to-source voltage ($V_{GS} < V_{TH}$, where V_{TH} is negative for a PMOS) is required to form a p-type inversion channel between the source and drain. Current then flows when a negative drain-to-source voltage (V_{DS}) is applied. Depending on the relative magnitudes of V_{GS} , V_{DS} and V_{TH} , the PMOS operates in one of three regions:

- Cut-off region: $V_{GS} > V_{TH}$ no channel is formed and the drain current $I_D \approx 0$.
- Triode / Linear region: $V_{GS} < V_{TH}$ and $V_{DS} > (V_{GS} - V_{TH})$ the device behaves like a voltage-controlled resistor and I_D increases roughly linearly with V_{DS} (in magnitude).
- Saturation region: $V_{GS} < V_{TH}$ and $V_{DS} \leq (V_{GS} - V_{TH})$ the channel is pinched off near the drain and I_D becomes almost independent of V_{DS} , depending mainly on V_{GS} .

The output (drain) characteristics of a PMOS are obtained by plotting I_D versus V_{DS} for several fixed, negative values of V_{GS} . Because both the gate and drain are biased negative with respect to the source, all voltages and currents in the resulting plots are negative; the current magnitude still increases with the magnitude of the gate overdrive voltage, exactly as in the NMOS case but mirrored about the origin.

3. Software / Tools Required

- Cadence Virtuoso Schematic Editor
- GPD090 Technology / Analog Design Kit
- Analog Design Environment (ADE-L) for DC and Parametric Analysis

5. Circuit Diagram

The test bench consists of a single PMOS transistor (gpdk090_pmos1v, $W = 120 \text{ nm}$, $L = 100 \text{ nm}$) whose Gate is biased through a DC voltage source VGS, whose Source is tied directly to ground (gnd), and whose Drain is biased through a second DC voltage source ($V_{DS} = -1.8 \text{ V}$) referenced to the same ground.

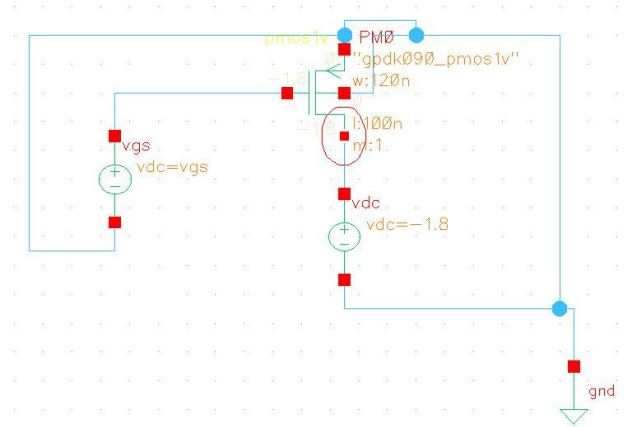


Fig. 2: Schematic of the PMOS DC characterization test bench built in Cadence Virtuoso

6. Procedure (Steps to Perform the Experiment)

6.1 Creating the Library and Schematic

1. Open Cadence Virtuoso. Go to File → New → Library, attach it to the GPDK090 technology library, and click OK.
2. Go to File → New → Cellview. Enter a suitable cell name (e.g., "Lab2_PMOS"), keep the tool as Schematic (Composer-Schematic), and click OK.
3. In the schematic window, go to Add → Instance (or the Add Instance icon). Click Browse and select the library "GPDK090".
4. From the cell list, select the cell "pmos1v" with View: symbol, and place it in the schematic window.
5. Add the Ground/Power symbols from the analogLib library — place a "gnd" symbol and two "vdc" (DC voltage source) symbols, one for VGS and one for VDS.
6. Wire the connections as shown in the circuit diagram (Fig. 1): Gate → VGS source, Source → gnd, Drain → VDS source (set to -1.8 V) → gnd.
7. Check the schematic for errors (Check and Save) and save the design.

6.2 Setting up the DC Analysis (ADE-L)

8. Launch the Analog Design Environment: Launch → ADE-L.
9. Go to Analyses → Choose and select the "dc" analysis type from the Analyses (AC / DC / Trans) options.
10. In the DC Analysis form, set Component Parameters: pick the Drain-side voltage source (VDS) as the component to sweep.
11. Set the sweep range: Start = -1.8 V , Stop = 0 V , with Sweep Type = Linear and Step Size = 0.2 V .
12. Under Outputs, select "To be plotted" and click on the desired node/terminal on the schematic (drain current through the transistor) to add it as an output — this generates the I_D vs V_{DS} plot (Graph 1).

13. Click on the DC Voltage field of the VGS source in the schematic to select it for the parametric sweep, and run the simulation to view the first output characteristic curve.

6.3 Parametric Analysis (Family of Curves for Different VGS)

14. Go to Tools → Parametric Analysis (Add Variables) in the ADE-L window.
15. Select the variable VGS, and set its Range = -1.8 V to 0 V, with Step Mode = Linear Steps and Step Size = 0.2 V.
16. Run the parametric sweep. This re-runs the DC analysis for each stepped value of VGS, producing a family of I_D - V_{DS} curves one curve per VGS value on the same graph (Graph 1).
17. Individual curves can also be isolated/viewed one at a time by toggling visibility in the plot legend (Graph 2).

7. Result / Observations

The DC output characteristics of the PMOS transistor were plotted with the drain current I_D (in μA) on the Y-axis and the drain-to-source voltage V_{DS} (in V) on the X-axis, for V_{GS} stepped from -1.8 V to 0 V in steps of 0.2 V. Both the voltage and current axes are negative, since the PMOS is biased and conducts with negative gate and drain voltages relative to the source.

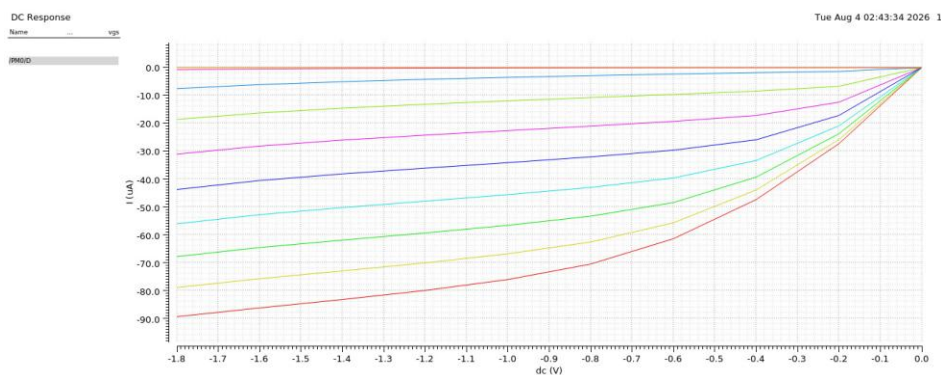


Fig. 3: Output characteristics (I_D vs V_{DS}) of the PMOS: family of curves for V_{GS} stepped from -1.8 V to 0 V (parametric analysis)

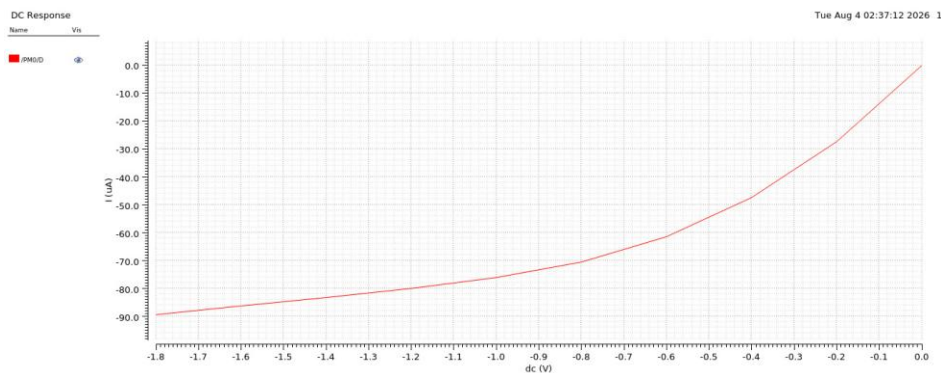


Fig. 4: Output characteristic (I_D vs V_{DS}) of the PMOS for a single fixed value of V_{GS} , isolated from the family of curves in Fig. 2

As seen in Fig. 2, each curve falls steeply (in magnitude) for V_{DS} close to 0 V (triode/linear region), then bends and flattens as V_{DS} becomes more negative, beyond $(V_{GS} - V_{TH})$, entering the saturation region where I_D becomes nearly constant. Curves corresponding to more negative V_{GS} values saturate at higher current magnitudes, confirming the expected dependence of saturation current on the gate overdrive voltage mirroring the NMOS behavior but with reversed polarity.

8. Conclusion

The DC (output) characteristics of a P-MOS transistor were successfully obtained in Cadence Virtuoso using the GPDK090 technology by sweeping V_{DS} from -1.8 V to 0 V through DC analysis and stepping V_{GS} from -1.8 V to 0 V through parametric analysis. The resulting family of I_D V_{DS} curves clearly shows the triode and saturation regions of operation, and demonstrates that the magnitude of the drain current increases with increasing magnitude of the gate-to-source voltage, consistent with standard MOSFET theory and mirroring the behavior observed for the NMOS in Experiment 1. This experiment provided hands-on familiarity with PMOS schematic capture, DC analysis setup, and parametric sweeps in the Cadence ADE-L environment.